Supplementary Material: The Anchor Accumulation Problem: Bounded Anchor Sets, a Soft–Hard Protection Law, and the Auditable Cost of Regularized Continual Learning

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A Hyperparameters and reproducibility

Statistics. All comparisons between two arms are *paired by seed*: each seed supplies one run per arm under an identical protocol, and every reported Δ is the per-seed difference. We report the paired t statistic $t(9)$ together with the two-sided p-value of the *Wilcoxon signed-rank* test on the same paired differences, and we report the Shapiro–Wilk p-value on those differences so that the normality assumption underlying the t test is visible rather than implicit. The choice to report both is deliberate. Whether a benchmark's headline number supports a claim depends on the variability of the metric under resampling, not only on its mean. Several authors have argued this point for generative evaluation [3, 4], for metric learning [5], for deep reinforcement learning [1], and, closest to our setting, for reproducibility in machine-learning scholarship as a whole [2]. A single significance test applied to ten seeds is exactly the setting in which those critiques bite: with ten paired samples the parametric t test has assumptions that the data may not satisfy, while the distribution-free signed-rank test is valid at this sample size by construction. We therefore report both and let the reader see where they diverge. The two tests agree on every headline comparison: the FIFO budget beats unbounded at M=2 and M=5, T=10 and T=20, with Wilcoxon p=0.002 (the smallest attainable two-sided value at n=10 paired samples) in all four cases. One case deserves the explicit note: at M=5, T=10 the paired differences depart from normality (Shapiro–Wilk p=0.024), so the t test there rests on an assumption the data do not satisfy, whereas the distribution-free Wilcoxon result (p=0.002) does not. We therefore treat the Wilcoxon value as the primary significance statement throughout, and the t statistic as a secondary, parametric cross-check. The significance threshold is α=0.05 throughout, with no multiplicity correction: the policy×budget grid of the Split-CIFAR-100 table in the main text contains twelve cells and is read as a descriptive map of the policy landscape, with per-cell p-values reported so that the reader can judge each claim individually. All p-values are recomputed from the raw per-seed records by `audit/audit_nonparametric.py`, which is included in the accompanying package.

Permuted-MNIST / Fashion-MNIST (main-text experiment sections). Architecture as in the audit [6]: a 784–256 fully connected trunk whose pre-activation is modulated by K=T dendritic segments (linear context-to-segment map, winning-segment readout), hidden activity sparsified by k-WTA with k=64 = $N/4$, linear 10-way head. Adam, lr 10^{-3}, batch 256, two epochs and 10,000 samples per task; permutation matrices drawn once per seed and shared by all arms. Diagonal Fisher and usage statistics from 300 and 500 samples per task respectively (max-log-probability Fisher variant); λ_{DeLP}=5000, corrected-SI reference λ=100. Paired arms use seeds 0–9; the policy

grid seeds 0–2 (policy grid table in the main text); the λ grid as in the λ-interaction table of the main text. Every run seeds `torch.manual_seed` before model construction. Code and per-seed JSON records (including `survivor_history` and `acc_matrix`) are released with the repository.

Split-CIFAR-100 (main text, Sec. on generalization). ResNet-18 adapted for 32×32 inputs (3×3 stem, stride 1, no max-pool) with all BatchNorm layers replaced by GroupNorm (32 groups). SGD, momentum 0.9, weight decay 5×10^{-4}, lr 0.05, batch 128, five epochs per task. Masked cross-entropy over the task's $100/T$ classes; task-incremental evaluation over the same classes. EWC penalty over backbone parameters only, λ=0.1; diagonal Fisher estimated from 50 single-sample draws of the squared gradient of the max log-softmax on the finished task, rescaled to unit global mean; gradient clipping at max norm 5. Tasks are contiguous $100/T$-class blocks of a fixed random permutation of the 100 classes (generator seed 100+seed, matching the released code); seeds 0–9 control init, shuffle, and eviction randomness. Per-job crash-safe checkpoints and the full JSON records (including `survivor_history` and `acc_matrix`) are released with the code.

B Full per-task accuracy matrices

Final-phase accuracy (%) per task, Permuted-MNIST, λ=5000, M=5: mean over seeds (n=10 for unbounded and fifo, n=3 for random/redun/merge). These underlie Fig. 3a and the grandfathering analysis of the main text.

Table 1: T=10.

task	unbounded	fifo	random	redun	merge
1	82.3	74.3	83.0	83.5	82.3
2	84.6	82.1	82.9	85.5	84.6
3	81.7	80.1	79.3	82.1	81.7
4	72.6	70.5	71.9	73.3	72.6
5	65.7	66.2	62.5	66.3	65.7
6	52.7	53.8	38.0	18.7	52.7
7	36.1	41.9	49.2	22.4	36.0
8	29.3	44.8	39.9	20.8	29.3
9	24.6	42.9	42.3	25.4	24.9
10	23.2	40.3	52.3	58.2	23.1

Data availability

All code, experiment scripts, and the complete JSON records of every run (including per-phase accuracy matrices and anchor-stack survivor histories) are publicly available at `https://github.com/zhouranxi/delp-anchor-budget`. This manuscript and its full reproducibility package — including the audit scripts that re-derive every quoted number from those records — are archived on Zenodo (doi:10.5281/zenodo.23668522). The precursor audit study is archived at doi:10.5281/zenodo.23572226 and the companion segment-specialization audit at doi:10.5281/zenodo.23637972; the underlying dataset and code release is doi:10.5281/zenodo.23639561.

Table 2: T=20.

task	unbounded	fifo	random	redun	merge
1	80.6	67.4	70.7	84.4	80.6
2	83.8	73.3	78.6	83.4	83.8
3	79.6	68.9	69.6	78.3	79.7
4	70.8	59.2	56.4	69.3	70.5
5	60.6	52.8	48.4	61.1	60.6
6	52.5	41.8	46.1	11.6	51.8
7	36.7	34.6	38.1	16.3	36.6
8	30.5	38.1	31.7	16.9	30.2
9	26.2	37.9	36.7	17.4	26.2
10	17.5	29.9	35.4	13.5	17.4
11	14.5	31.8	21.4	14.8	14.4
12	10.9	40.3	30.7	14.9	11.0
13	14.8	32.4	28.1	19.4	14.7
14	11.8	34.4	28.8	20.2	11.6
15	16.2	36.7	30.0	25.1	16.2
16	18.3	39.5	42.3	30.9	18.5
17	14.1	41.3	38.0	23.7	13.9
18	9.8	35.4	31.2	30.4	9.7
19	9.6	30.4	36.0	35.0	9.3
20	14.6	41.7	41.2	60.1	14.4

References

[1] Rishabh Agarwal, Max Schwarzer, Pablo Samuel Castro, Aaron Courville, and Marc G. Bellemare. Deep reinforcement learning at the edge of the statistical precipice. In *Advances in Neural Information Processing Systems*, volume 34, 2021.

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[6] Ranxi Zhou. Where do the gains come from? An audit of dendritic context-gated protection for continual learning. Preprint, Zenodo, doi:10.5281/zenodo.23572226, 2026. Code: `https://github.com/zhouranxi/delp-audit`.